

2023 AMC 10A Problem 16 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 16

Problem:

In a table tennis tournament every participant played every other participant exactly once. Although there were twice as many right-handed players as left-handed players, the number of games won by left-handed players was 40% more than the number of games won by right-handed players (there were no ties and no ambidextrous players). What is the total number of games played?

Answer Choices:

- A. 15
- B. 36
- C. 45
- D. 48
- E. 66

Solution:

Let ℓ denote the number of left-handed players, and let r denote the number of righthanded players. There were $\frac{\ell(\ell-1)}{2}$ games between two left-handed players, $\frac{r(r-1)}{2}$ games between two right-handed players, and $\ell \cdot r$ games involving one left-handed and one right-handed player. Let n denote the number of games won by right-handed players against left-handed players. Then righthanded players won a total of $\frac{r(r-1)}{2} + n$ games, and left-handed players won $\frac{\ell(\ell-1)}{2} + (\ell \cdot r - n)$ games. Because left-handed players won 40% more games than right-handed players,

$$\frac{\ell(\ell-1)}{2} + \ell \cdot r - n = 1.4 \left(\frac{r(r-1)}{2} + n \right)$$

Using $r = 2\ell$ and simplifying this equation gives $\ell^2 - 3\ell + 8n = 0$. If $n \geq 1$, there are no real solutions for this quadratic equation. Therefore $n = 0$, giving $\ell = 3$ and $r = 6$. The total number of games played is

$$\frac{\ell(\ell-1)}{2} + \frac{r(r-1)}{2} + \ell \cdot r = 3 + 15 + 18 = (\text{B}) \boxed{36}$$

Note that the $r = 6$ right-handed players won a total of 15 games (all among themselves), and the $\ell = 3$ left-handed players won a total of 21 games (3 games among themselves and all 18 games against right-handed players), and 21 is indeed 40% more than 15.

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